

Joaquim Ortiz-Haro

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Citizenship: Spanish

I am a Research Scientist at Helsing AI (formerly Keybotic) in Barcelona, where I work on deep reinforcement learning for robotic locomotion. Previously, I was a Postdoctoral Researcher in Robotics at New York University (NYU) in the Machines in Motion Laboratory, supervised by Prof. Ludovic Righetti. My research interests include locomotion, task and motion planning, control, and deep learning for robotics. I received my PhD in Robotics from TU Berlin, supervised by Prof. Marc Toussaint.

EDUCATION

- **PhD in Robotics** Sep. 2019 - Jan. 2024
TU Berlin (Germany) and IMPRS-IS (Max Planck Research School). Focus on task and motion planning, nonlinear optimization, and deep learning for robotics. Advisor: Marc Toussaint. Graduated summa cum laude (maximum qualification).
- **Master of Mathematics** 2017 - 2019
UPC (Polytechnic University of Catalonia, Barcelona), Faculty FME. Average qualification: 9.38/10. Exchange student at LAAS-CNRS (French National Research Center).
- **Bachelor of Mechanical Engineering** 2013 - 2017
UPC (Polytechnic University of Catalonia, Barcelona), Faculty ETSEIB. Top Class Honours. Ranked 2nd out of 350 graduate students. Average qualification: 8.84/10. Exchange student at ETH Zurich (Switzerland).
- **High School** 2014 - 2017
Honour Mention in the Spanish University Entrance Exam (13.36/14). Scholarship to study the first year at university. Gold Medal in the XXVI Chemistry Spanish National Olympiad.

WORK EXPERIENCE

- **Robotics and AI Research Scientist – Helsing AI/Keybotic** Jan. 2025 - Ongoing
Main research project: state-of-the-art vision-based locomotion for legged robots. Trained RL policies, developed new architectures and training pipelines, conducted sim-to-sim evaluation and sim-to-real transfer, and deployed policies on real robots.
- **Postdoctoral Researcher, Robotics, NYU** Feb. 2024 - Jan. 2025
Machines in Motion Laboratory, New York University (NYU), under the direction of Professor Ludovic Righetti. Conducting research in task and motion planning, control, and deep learning for robotics. Leading research projects, supervising PhD and Master's students, and collaborating with team members on various research projects.
- **PhD Student, Robotics** Sep. 2019 - Jan. 2024
See section Education.
- **Research Intern, Machine Learning** 2019
Worked on optimal transport and theoretical machine learning. Artificial Intelligence and Machine Learning Group at UPF (University Pompeu Fabra, Barcelona). Supervisor: Gergely Neu.
- **Research Intern, Robotics** 2018
Conducted research on contact and trajectory optimization using an augmented Lagrangian algorithm. Humanoid Robotics at LAAS-CNRS (French National Research Center). Supervisor: Nicolas Mansard.
- **Intern, Robotics Engineer** 2018
Performed geometric and semantic analysis of 3D point clouds. Scaled Robotics (start-up, Barcelona).

- **Research Assistant, SLAM** 2017
Developed parallel tracking and mapping algorithms for an event-based camera. IRI (Robotics Institute, Barcelona). Supervisor: Joan Solà.
- **Teaching Assistant** 2015 - 2017
Taught mathematics to first-year engineering students at ETSEIB, UPC.
- **Private Academy Teacher** 2014
Taught university-level engineering lessons at a private academy.

SELECTED PUBLICATIONS

Google Scholar Profile: [Joaquim Ortiz-Haro](#). Check also my personal [website](#). Citations: 367, h-index: 10 (Google Scholar, September 2026).

1. WorldPlanner: Monte Carlo Tree Search and MPC with Action-Conditioned Visual World Models. R. Khorrambakht, [Joaquim Ortiz-Haro](#), Joseph Amigo, Omar Mostafa, Daniel Dugas, Franziska Meier, and Ludovic Righetti. IEEE International Conference on Robotics and Automation (ICRA), 2026.
2. db-ECBS: Interaction-Aware Multirobot Kinodynamic Motion Planning. Akmaral Moldagalieva, [Joaquim Ortiz-Haro](#), and Wolfgang Hönig. IEEE Transactions on Robotics (T-RO), 2026.
3. Factored Task and Motion Planning with Combined Optimization, Sampling, and Learning. [Joaquim Ortiz-Haro](#). PhD Thesis, TU-Berlin, 2024.
4. iDb-A*: Iterative Search and Optimization for Optimal Kinodynamic Motion Planning. [Joaquim Ortiz-Haro](#), Wolfgang Hönig, Valentin N. Hartmann, and Marc Toussaint. IEEE Transactions on Robotics (T-RO), 2024.
5. Conflict-driven Interface between Symbolic Planning and Nonlinear Constraint Solving. [J. Ortiz-Haro](#), E. Karpas, M. Katz, and M. Toussaint. IEEE Robotics and Automation Letters (RA-L), 2022.
6. Structured Deep Generative Models for Sampling on Constraint Manifolds in Sequential Manipulation. [Joaquim Ortiz-Haro](#), Jung-Su Ha, Danny Driess, and Marc Toussaint. 5th Annual Conference on Robot Learning (CoRL), 2021.

PROGRAMMING

- **C++**: Main tool for robotics research during PhD and Postdoc. Utilized in internships at robotics companies and robotics competitions. Proficient with BOOST, ROS, PCL, OpenCV, EIGEN, OMPL, and developed custom C++ libraries with Python bindings. Deployment of RL policies on quadruped robots.
- **Python**: Used for deep learning research during PhD (PyTorch). Applied in master's thesis for robotics and numerical optimization (SciPy, NumPy, etc.) and various programming courses. Built RL training and evaluation framework with Isaac Lab/Sim.
- **Matlab**: Utilized in bachelor thesis (robotics and computer vision) and university courses.
- **Other**: Experienced with Arduino, PlatformIO, ANSYS, SolidWorks, Minitab, and Autodesk Fusion.

LANGUAGES

- **Spanish and Catalan:** Native.
- **English:** Professional proficiency. Scientific publications, bachelor's and master's theses written in English. Advanced Certificate (CAE) from Cambridge University, C1.2 (2013).
- **French:** Official certificate DELF B2 (2019). Six-month exchange in Toulouse.
- **German:** Official certificate B2.2 from Goethe Institute (2017). Engineering courses conducted in German at ETH. Attended two summer language courses in Germany.

PERSONAL SKILLS

Passionate about research, problem-solving, and learning. Experienced in interdisciplinary and international teams. Dedicated, honest, and hardworking. I love teaching and sharing my knowledge. In my free time, I enjoy climbing, cycling, and reading books. I also play the guitar and am learning the piano.

STUDENT SUPERVISION

1. Real-time Task and Motion Planning and Control. Alexandre Gllr. Master's research internship (NYU).
2. Efficient Kinodynamic Motion Planning with Reinforcement Learning Policies. Alexander Weingart. Co-supervised with Wolfgang Hoenig. Master's Thesis in Computer Science (TU Berlin).
3. Neural Scene Representations for Sequential Reasoning. Phillip Groete. Co-supervised with Ozgur Oguz. Master's Thesis in Computer Science (TU Berlin).
4. SCP and k-Order Motion Optimization for Cooperative Multirotor Teams. Welf Rehberg. Co-supervised with Wolfgang Hoenig. Master's Thesis in Computer Science (TU Berlin).
5. Combining Sampling and Optimization for Optimal Motion Planning. Jay Kamat. Co-supervised with Andreas Orthey. Master's Thesis in Mathematics (BITS and TU Berlin).
6. Constrained Sampling - A Study on Methods for Sampling from Constraint Manifolds. Philip Oedi. Co-supervised with Jung-Su Ha. Master's Thesis in Computer Science (TU Berlin).

TEACHING

- **AI & Robotics: Lab Course.** M.Sc. Computer Science, TU-Berlin. Teaching Assistant, Summer Semester 2022.
- **Optimization Algorithms.** M.Sc. Computer Science, TU-Berlin. Teaching Assistant, Winter Semesters 2020, 2021, 2022, 2023.
- **Calculus II.** B.Sc. Industrial (Mechanical) Engineering, UPC Barcelona (ETSEIB). Teacher in Aula Lliure, Summer Semesters 2015, 2016, and 2017.

- **Calculus I.** B.Sc. Industrial (Mechanical) Engineering, UPC Barcelona (ETSEIB). Teacher in Aula Lliure, Winter Semester 2015.

INVITED TALKS

- **A Bidirectional Interface between Nonlinear Optimization and Symbolic Planning for Task and Motion Planning**
 - Seminar, NYU Robotics. Host: Ludovic Righetti. 2023.
 - INRIA Paris. Host: Justin Carpentier. 2023.
 - IRI (Robotics Institute, Barcelona). Host: Joan Sola. 2023.
 - University Pompeu Fabra (UPF). Host: Gergely Neu. 2023.
 - Robotics Seminar, Technion (Israel). Host: Erez Karpas. 2022.
- **Structured Deep Generative Models for Sampling on Constraint Manifolds in Sequential Manipulation**
 - Seminar of Group Gepetto, LAAS-CNRS (France). Host: Nicolas Mansard. 2021.

SERVICE

Reviewer: ICRA, IROS, RAL, CoRL, JAIR.

Last updated: September 2026.